

DATA ROLES

A Learning Journal

Notes on how Data Analysts, Data Engineers, Data Scientists and AI/ML Engineers actually work — and where they overlap.

What Data Is — and Who Works With It

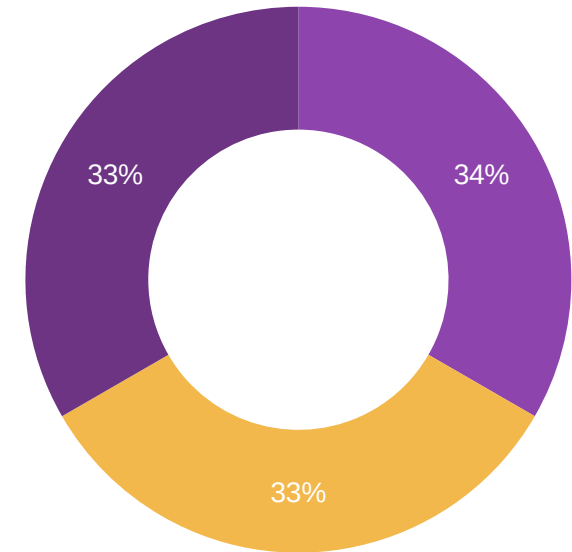
D What is Data?

- Information about different things gathered at one place — or about multiple things across multiple places
- Can be numeric, structured, or unstructured
- “Data is gold.”

H The Data Role Hierarchy

- Data Analyst → Data Engineer → Data Scientist → AI/ML Engineer
- Data Analyst: grabs and works the data to give benefit / value to the organization — in short, to analyze
- AI/ML Engineer path connects to Deep Learning

Forms Data Can Take



■ Numeric ■ Structured ■ Unstructured

Every dataset in these notes falls into one of these three buckets.

Tools of the Trade & Reading a Domain

T Tools for Data Analysis

- Excel
- Python (or R)
- SQL
- Power BI

R Reading the Domain & the Core Roles

- Check if the data is structured or not, and what the task actually needs
- Is it in attribute form? Is it laid out vertically or horizontally?
- Data Analyst role: reads existing data, builds reports & dashboards, supports business decisions — using SQL, Excel, BI tools

The Analyst's Toolkit



Domain first, tool second — the data decides the approach.

Engineer, Scientist, AI/ML — and the DA Skillset

R Data Engineer · Scientist · AI/ML Engineer

- Data Engineer: Structures data so it flows clearly to everyone else — Python, Spark, Cloud
- Data Scientist: Uses statistics & ML to predict, forecast and run experiments — Stats, ML, Python
- AI/ML Engineer: Deploys data that's been through the pipeline; deploys & scales ML models into production-ready products

S Key Skills for a Data Analyst

- SQL for querying data · Excel & spreadsheet formulas
- Power BI & Tableau dashboards · Basic Python & statistics
- Path in: any degree + Excel, SQL and BI tools

Data in the Real World

F The Focus of Each Role

- Data Analyst: business + reporting focus
- Data Engineer: software + infrastructure focus
- Data Scientist: statistics + ML focus
- AI/ML: ML + engineering focus — common base: SQL, Python, Statistics, Excel

W Data in the Wild

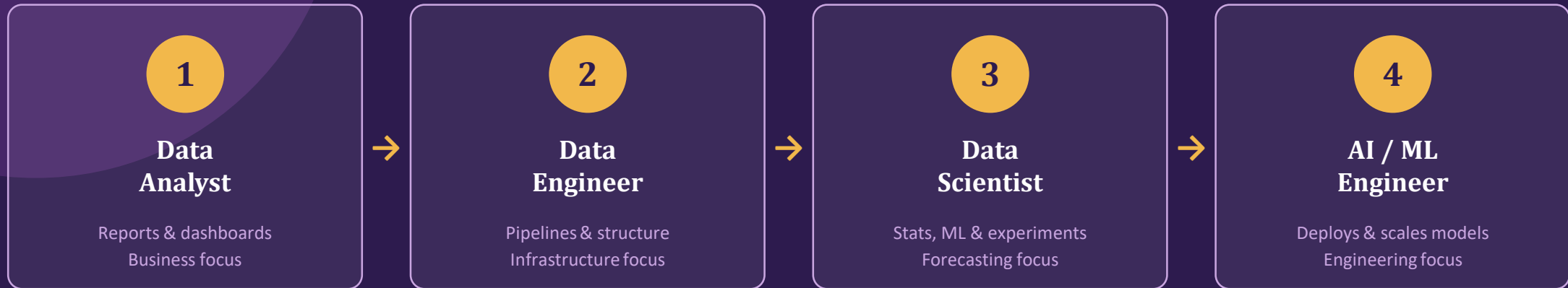
- Every domain has its own data analyst — e.g. social media, logistics
- Real examples: Amazon, Instagram, WhatsApp, Facebook — the “BTS of the uploading”
- To dig into a role: check responsibilities, teammates, type of work, domain and industry

Platforms Referenced Today



Every platform runs on a data analyst behind the scenes.

The Data Role Hierarchy



Common Ground

SQL · Python

Statistics · Excel

Whatever the role, the job is the same at its core: turn raw data into something the organization can trust.